Technical Report

Sound-Based Spin Estimation in Table Tennis: Reproduction and Analysis

Alex Zanetti[a,1]

[a] *University of Trento*

Abstract—This report reproduces and extends the sound-based table tennis spin estimation pipeline proposed by Gossard et al., which combines an energy-based bounce detector with a CNN classifier on 15 ms audio clips. Starting from the original pipeline, we compare multiple detection algorithms — including energy-based methods and new spectral methods (spectral flux, SuperFlux) — and evaluate their performance. The aim is to analyze how spin types manifest in the frequency domain, visualize feature separability via t-SNE, evaluate CNN classifiers for surface (13 classes) and spin direction (3 classes), and propose an improved detector with a pre-peak onset filter to reduce false positives.

1. INTRODUCTION

Spin is one of the most important factors in table tennis, directly affecting ball trajectory and making return shots significantly harder for the opponent. Current spin estimation methods rely on high-speed cameras and computer vision, which are expensive and impractical for amateur settings.

Gossard et al. [3] proposed an alternative approach: estimating spin from the *sound* of ball bounces. Their pipeline combines energy-based peak detection with a convolutional neural network (CNN) classifier operating on Mel spectrogram features extracted from 15 ms bounce clips.

This report reproduces and extends their work by comparing six detection algorithms (four energy-based and two spectral), analyzing how spin types manifest in the frequency domain using t-SNE [1] embeddings, and evaluating CNN classifiers for surface type and spin direction.

2. DATASET

The dataset [4] consists of annotated bounces recorded with a Zoom H4n Pro directional microphone at 44.1 kHz

- **10 racket configurations** varying blade type, sponge thickness, and rubber.
- **Simple vertical bounces** (files `01.wav–10.wav`): clean recordings with minimal noise.
- **Serve recordings** (`backhand-*.wav`, `forehand-*.wav`): real game-like conditions where the racket stroke precedes the bounce.
- **Table and floor bounces**: distinct surface types for classification evaluation.
- **"Other" sounds** (`caps.wav`, `pen.wav`, etc.): non-bounce sounds for negative examples.

The dataset contains 3,396 racket bounces, 777 table bounces, 290 floor bounces, and 1,239 other sounds, split 80/20 into training and test sets. Individual bounce clips are 15 ms long (661 samples). Fig. 1 shows the distribution of surface types and spin directions.

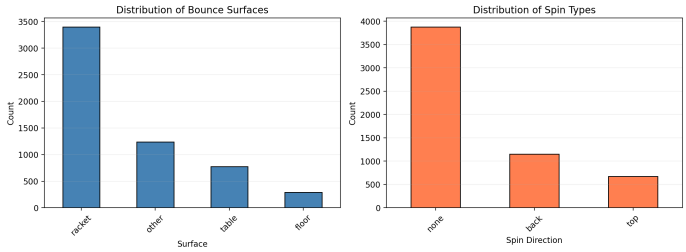

Figure 1. Distribution of surface types (left) and spin directions (right). Topspin has fewer samples, motivating the use of weighted loss.

3. SIGNAL PROCESSING PIPELINE

3.1. High-Pass Filtering

Table tennis ball bounces concentrate energy at **11 kHz and above**, while noise sources like speech and footsteps sit at lower frequencies. A **5th-order Butterworth high-pass filter** with cutoff $f_c = 10$ kHz removes this noise:

$$|H(j\omega)|^2 = \frac{1}{1 + \left(\frac{\omega}{\omega_c}\right)^{2N}} \tag{1}$$

where $\omega_c = 2\pi f_c$ and $N = 5$. Zero-phase filtering (`sosfiltfilt`) is used to avoid shifting transients in time, which matters at millisecond-level detection.

Fig. 2 shows the effect on `01.wav`: bounce transients remain intact while low-frequency noise is removed, making energy peaks clearly distinguishable.

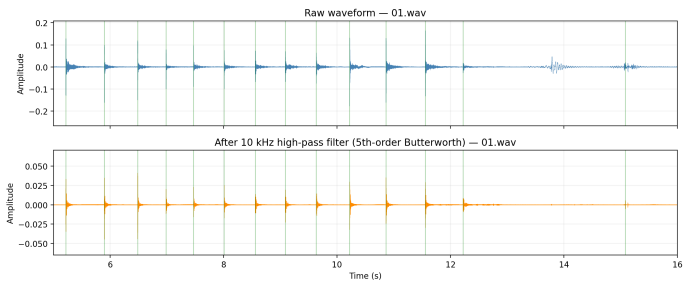

Figure 2. Waveform of `01.wav` before (top) and after (bottom) 10 kHz high-pass filtering. Green lines mark ground truth bounce timestamps. Bounce peaks are preserved while ambient noise is suppressed.

3.2. Frame Energy Computation

The filtered signal is split into non-overlapping **1 ms frames** ($L = 44$ samples at 44.1 kHz). For each frame k, the mean absolute energy is:

$$E_k = \frac{1}{L} \sum_{i=1}^{L} |x_{k,i}| \tag{2}$$

This yields a 1 kHz energy envelope, reducing cost while keeping enough temporal resolution for onset detection.

Based on the paper by T. Gossard, J. Schmalzl, A. Ziegler, and A. Zell: "Sound-Based Spin Estimation in Table Tennis: Dataset and Real-Time Classification Pipeline" [3].

3.3. Exponential Decay Average Detector

The main detector from [3] uses an **exponential moving average** (EMA) as an adaptive threshold. The decay average D_k is updated for each frame:

$$D_k = \gamma \cdot D_{k-1} + (1-\gamma) \cdot E_k \tag{3}$$

A bounce is detected when two conditions are met:

$$E_k \geq \beta \cdot D_k \quad \text{and} \quad k - k_{\text{last}} \geq T_{\text{timeout}} \tag{4}$$

where γ controls the decay rate, β sets the detection threshold, and $T_{\text{timeout}} = 100\,\text{ms}$ prevents repeated detections of the same bounce.

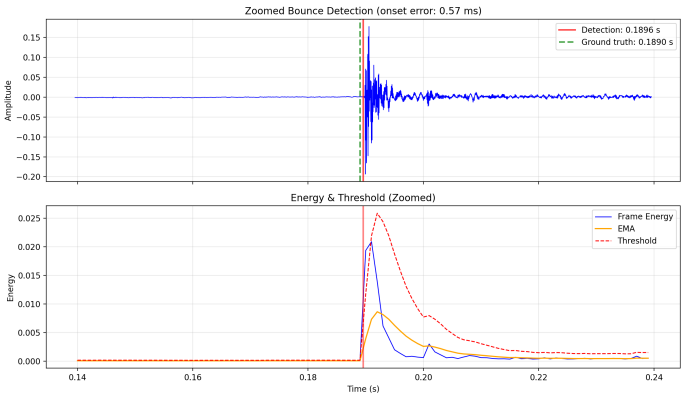

Figure 3. Zoomed bounce detection on `01.wav`. The detector fires 0.57 ms after the ground truth onset, well within the 5 ms strict tolerance.

3.4. Pre-Peak Onset Filter

Real bounces tend to emerge from relative silence, producing a sharp onset, while false positives often occur during sustained energy (e.g., mid-stroke). To capture this, we define the **pre-peak ratio** as the detected frame's energy divided by the mean energy of the W preceding frames:

$$R_k = \frac{E_k}{\frac{1}{W}\sum_{j=k-W}^{k-1} E_j} \tag{5}$$

A candidate peak is accepted only if $R_k \geq R_{\min}$. Fig. 4 illustrates the two cases: a true bounce produces a large ratio (spike from silence), while a false positive during sustained energy yields a low ratio and is rejected.

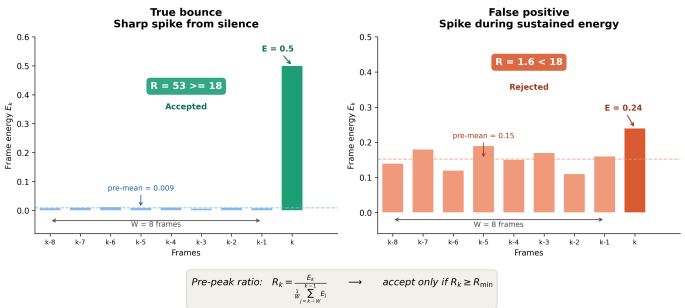

Figure 4. Pre-peak ratio for a true bounce (left, $R = 53$, accepted) vs. a false positive (right, $R = 1.6$, rejected). Real bounces spike sharply from near-zero energy; false positives rise from an already elevated baseline.

The final detection condition with the pre-peak filter becomes:

$$E_k \geq \beta \cdot D_k \quad \text{and} \quad R_k \geq R_{\min} \quad \text{and} \quad k - k_{\text{last}} \geq T_{\text{timeout}} \tag{6}$$

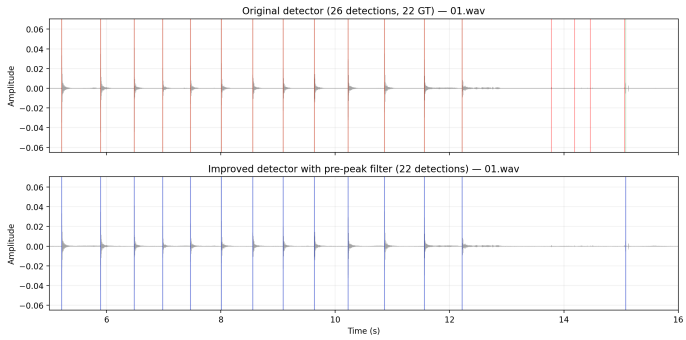

Figure 5. Detection output on `01.wav`. Top: original detector (more false positives). Bottom: improved detector with pre-peak filter ($R_{\min} = 18$, $W = 8$). Green lines: ground truth.

4. DETECTION ALGORITHM COMPARISON

Beyond the EMA-based approach, we compared multiple detection strategies to evaluate trade-offs between methods.

4.1. Energy-Based Detectors

We compare four energy-based approaches, all operating on the high-pass filtered signal:

1. **Adjacent Frame**: compares consecutive frames against a fixed threshold; simplest baseline.
2. **Moving Average**: uses the last 100 frames as reference; stable but slow to adapt.
3. **Decay Average (EMA)**: the approach from [3]; adapts continuously to the local energy level.
4. **Scipy Peaks**: applies `scipy.signal.find_peaks` with a prominence threshold on the energy curve.

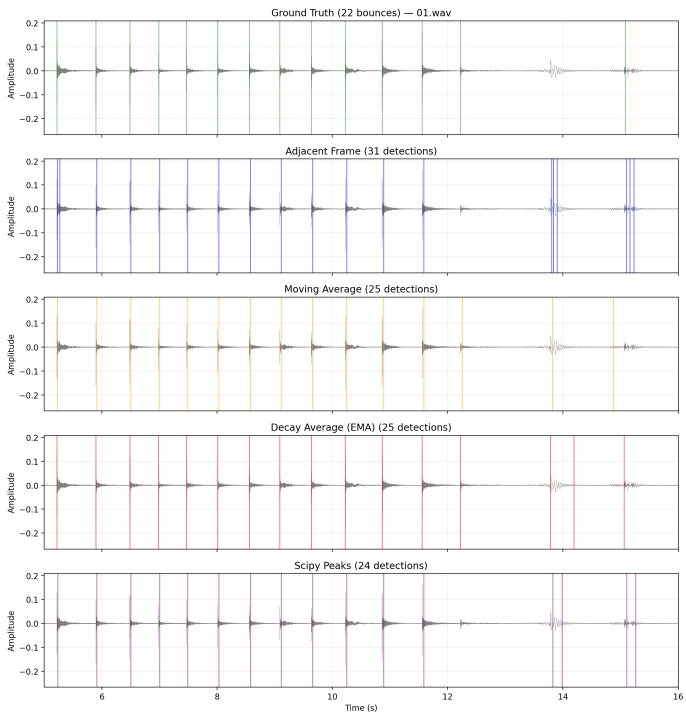

Figure 6. Comparison of four energy-based detectors on `01.wav`. Top panel: ground truth. Subsequent panels: detections from each method. The detection count for each method is shown in the title.

4.2. Spectral Detectors

We also evaluate two spectral onset detectors:

- **Spectral Flux** [5]: measures the half-wave-rectified frame-to-frame increase in STFT magnitude.
- **SuperFlux** [2]: extends spectral flux with a maximum filter along the frequency axis to suppress vibrato and slow modulations.

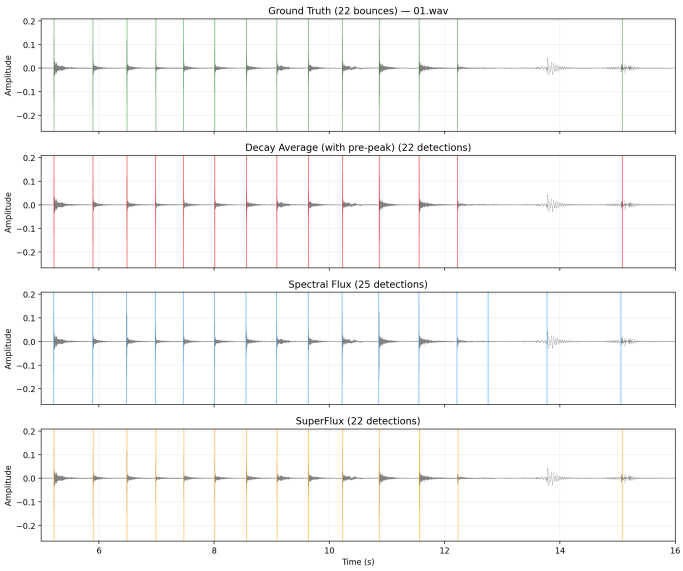

Figure 7. Spectral detectors vs. Decay Average baseline on `01.wav`. Spectral methods tend to detect onsets slightly before the annotated peak, affecting strict-tolerance metrics.

4.3. Quantitative Evaluation

Detectors were evaluated across multiple audio files at 5 ms (strict) and 20 ms (relaxed) tolerance windows. At strict tolerance, the Decay Average with pre-peak filter performs best, as its timestamps align closely with annotated peaks. At relaxed tolerance, spectral methods become competitive since their earlier onsets fall within the wider window.

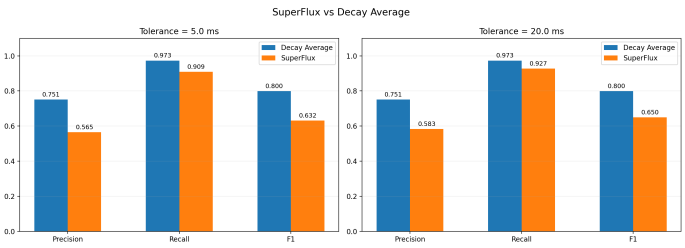

Figure 8. Precision, recall, and F1 for Decay Average vs. SuperFlux at 5 ms (left) and 20 ms (right) tolerance.

5. FEATURE EXTRACTION AND BOUNCE ANALYSIS

5.1. Bounce Analysis

Each 15 ms bounce clip (661 samples at 44.1 kHz) is examined through three representations:

- **Time-domain waveform**: shows the impulsive nature of bounces and visible shape differences across surfaces and spins.
- **Mel spectrogram**: FFT size 1024, window 128, hop 64, 64 Mel bands, 80 dB range, producing a $1 \times 64 \times 10$ feature map.
- **MFCC**: 40 Mel-frequency cepstral coefficients, capturing the spectral envelope compactly.

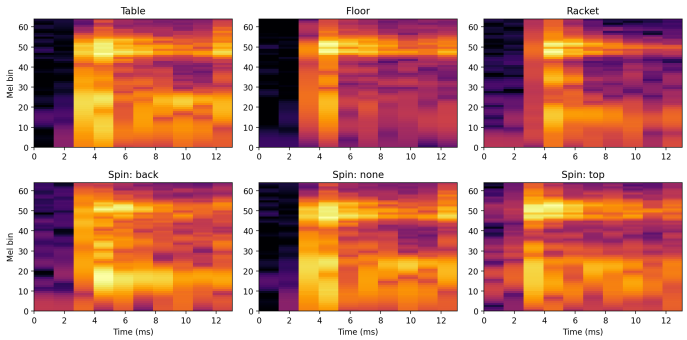

Figure 9. Mel spectrograms for different surfaces (top) and spin types (bottom). Surface differences are clear; spin differences are subtler.

5.2. Spin Frequency Analysis

Gossard et al. [3] show that spin presence can be detected from sound, though spin direction is harder to distinguish. To investigate, we compare average frequency spectra of 100 randomly sampled clips per spin type.

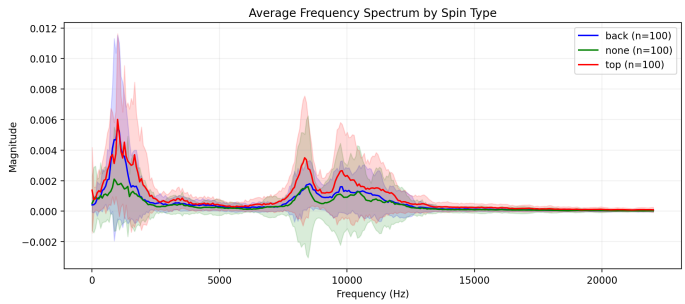

Figure 10. Average frequency spectra by spin type (100 samples each, shaded: ±1 std). Spin-present bounces show a clear peak around 1–2 kHz absent in no-spin. The overlap between backspin and topspin explains why direction classification is harder.

6. CLASSIFICATION

6.1. CNN Architecture

The classifier is a 6-layer CNN operating on Mel spectrogram inputs of shape $1 \times 64 \times 10$. Table 1 details the architecture. Each convolutional layer is followed by ReLU activation and batch normalization. Weights are initialized with Kaiming normal initialization. The model has ~25,000 parameters and is trained with Adam (lr $= 10^{-3}$) and cross-entropy loss. For spin classification, class-weighted cross-entropy is used to handle the imbalanced distribution.

Table 1. CNN architecture. All conv layers include ReLU + BatchNorm.

Layer	Kernel	Stride	Channels	Output
Input	–	–	1	64×10
Conv1 + BN	5×5	$(2, 2)$	2	32×5
Conv2 + BN	3×3	$(2, 1)$	4	16×5
Conv3 + BN	3×3	$(2, 1)$	8	8×5
Conv4 + BN	3×3	$(2, 1)$	16	4×5
Conv5 + BN	3×3	$(2, 1)$	32	2×5
Conv6 + BN	3×3	$(2, 1)$	64	1×5
AdaptiveAvgPool	–	–	64	1×1
Linear	–	–	N	–

6.2. Surface Classification

The surface classifier distinguishes 13 classes (10 racket configurations + table + floor + other). Racket types with similar rubber properties tend to be confused, while structurally different surfaces (table, floor) are well-separated.

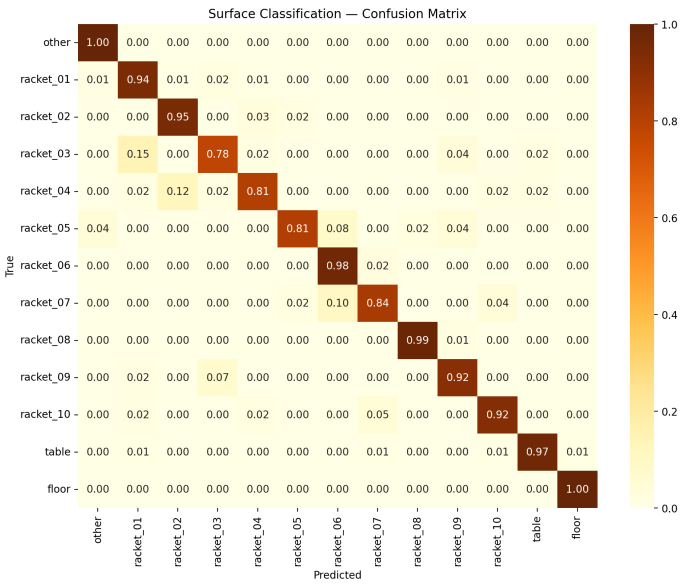

Figure 11. Normalized confusion matrix for surface classification (13 classes).

6.3. Spin Classification

The spin classifier operates on the same Mel spectrogram features to distinguish backspin, no-spin, and topspin. As reported by Gossard et al. [3], the main confusions occur between topspin and backspin, reflecting their acoustic similarity. Additionally, some spin samples may be misclassified as no-spin due to weak spin application that produces insufficient acoustic cues despite being labelled as spin — a form of label noise in the dataset.

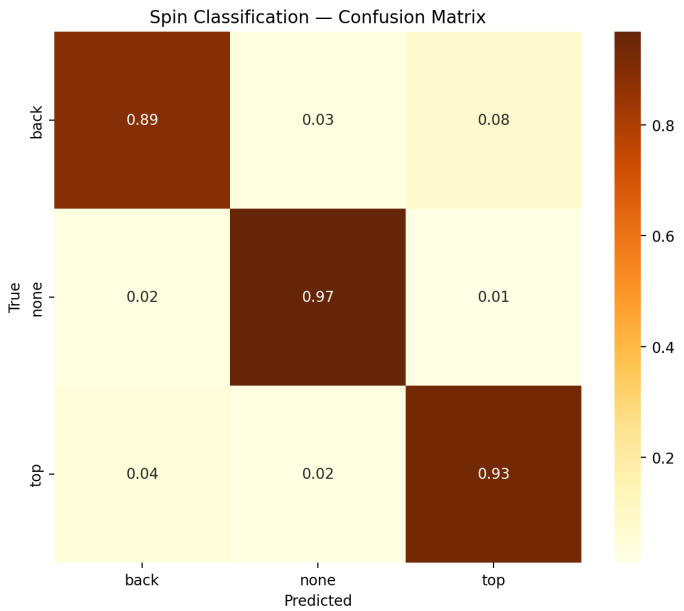

Figure 12. Normalized confusion matrix for spin classification (3 classes). Backspin–topspin confusion is highest, consistent with the spectral overlap observed in Section 3.4. Some spin samples fall near no-spin, likely due to weak spin application.

7. T-SNE VISUALIZATION

To verify that bounce sounds are distinguishable in feature space, we apply **t-SNE** [1] to normalized Mel spectrogram features from the test set. Each spectrogram is flattened into a feature vector, normalized, and projected to 2D.

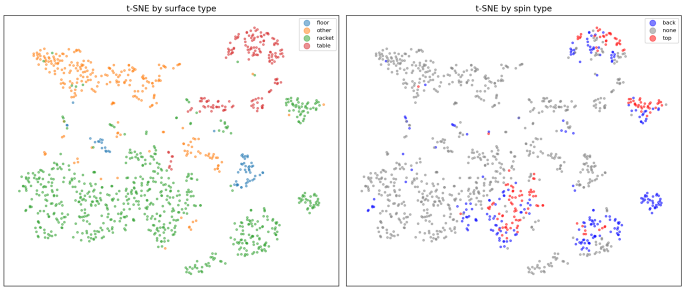

Figure 13. t-SNE visualization of Mel spectrogram features from the test set. Left: colored by surface type (racket types grouped). Right: colored by spin type. Surface clusters are well-separated; spin shows more overlap between back and top.

8. CONCLUSION

This report reproduced and extended the sound-based table tennis spin estimation pipeline by Gossard et al. [3]. The main findings are:

- **High-pass filtering at 10 kHz** effectively isolates bounce signatures, confirming that ball impacts concentrate energy above 11 kHz.
- Among **six detection algorithms** compared, the EMA detector offers the best balance of adaptivity and precision. Spectral methods (Spectral Flux, SuperFlux) tend to detect onsets earlier, affecting strict-tolerance metrics.
- A **pre-peak onset filter** improves the EMA detector by rejecting false positives during sustained energy, without requiring changes to the core EMA parameters.
- **Spin presence is acoustically distinguishable** from no-spin (stronger 1–2 kHz and 7–12 kHz components), but **spin direction remains harder to separate**, consistent with both spectral overlap and potential label noise from weak spin application.
- **t-SNE and CNN classifiers** confirm that surface types are well-separated in Mel spectrogram feature space, while spin clusters overlap more — particularly between backspin and topspin.

Future Work

- An **adaptive pre-peak threshold** that adjusts $R_{\min}$ to the local noise floor would make the filter robust beyond clean, controlled recordings.
- A broader dataset with **multiple players, rackets, and environments** — paired with **video-based spin verification** (e.g., ball logo tracking) — would improve label accuracy and generalization.

REFERENCES

[1] L. van der Maaten and G. Hinton, "Visualizing data using t-SNE", *Journal of Machine Learning Research*, vol. 9, 2008.

[2] S. Böck and G. Widmer, "Maximum filter vibrato suppression for onset detection", in *Proceedings of the 16th International Conference on Digital Audio Effects (DAFx-13)*, Maynooth, Ireland, Sep. 2013, pp. 55–61.

[3] T. Gossard, J. Schmalzl, A. Ziegler, and A. Zell, "Sound-based spin estimation in table tennis: Dataset and real-time classification pipeline", *arXiv preprint arXiv:2409.11760*, 2024.

[4] T. Gossard, J. Schmalzl, A. Ziegler, and A. Zell, *TT Sounds dataset*, 2024. [Online]. Available: https://cloud.cs.uni-tuebingen.de/index.php/s/p3tw3EqE9csXoRn.

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